Reason. Plan. Act. Physical AI Workshop

Lecture 1b: Day 1 Conclusion

Sarah Keren

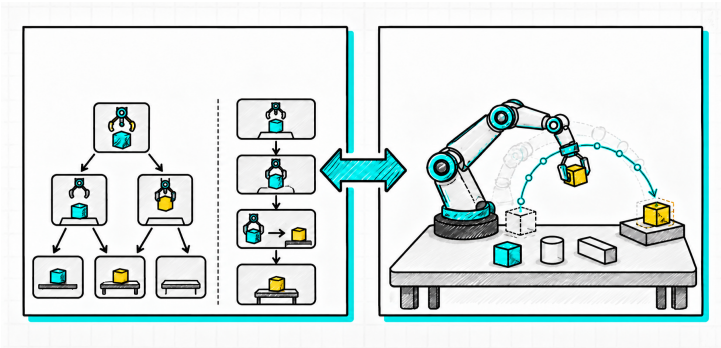
The Taub Faculty of Computer Science
Technion - Israel Institute of Technology



Recap

Physical AI

Provide a practical understanding of how Physical AI systems combine **discrete task decisions** with **continuous** geometry, collision **constraints**, and **feasible** robot motion.

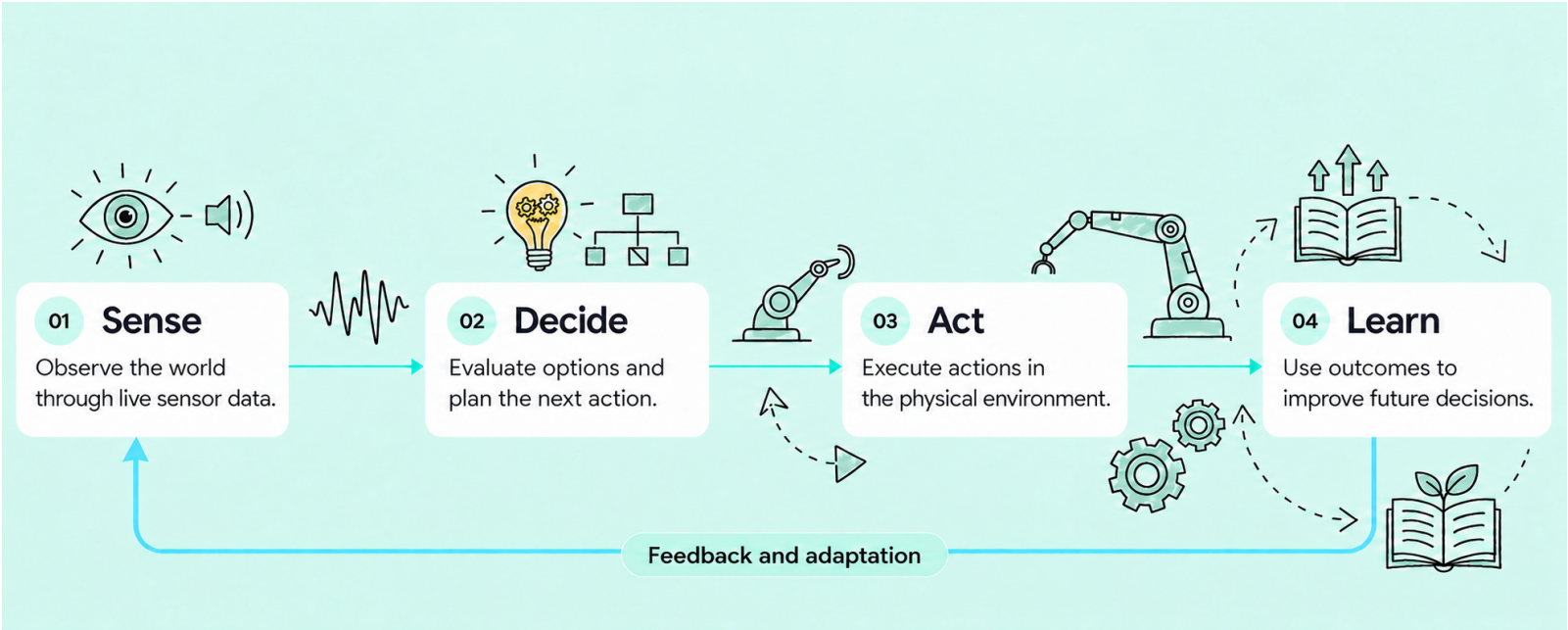


Account for **uncertainty and imperfect** sensing, changing environments, and execution **failures**.

Implement and evaluate core algorithms in simulation and on physical robotic platforms.

The Continuous Closed Loop

Continuous interaction between intelligence and the physical world.





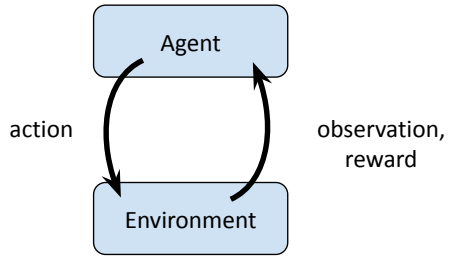
”Pick up an apple”

”I am hungry”

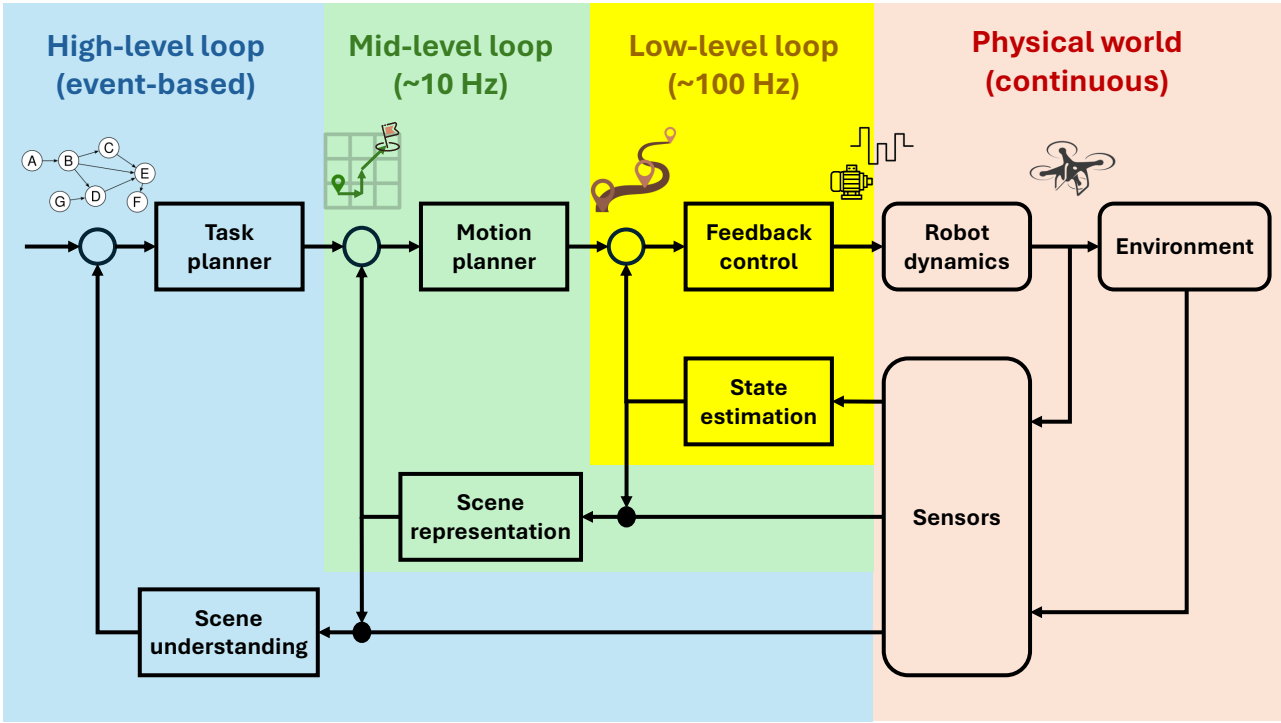
Autonomy

- Action (Actuation)
  - Locomotion
  - Manipulation
- Perception (Sensors)
  - Internal
  - External
- Cognition (Control)
  - From reactive to proactive
  - From finite state machines to cognitive robotics





Layered Decision-Making



Focus on state estimation and decision-making (and their relationship).

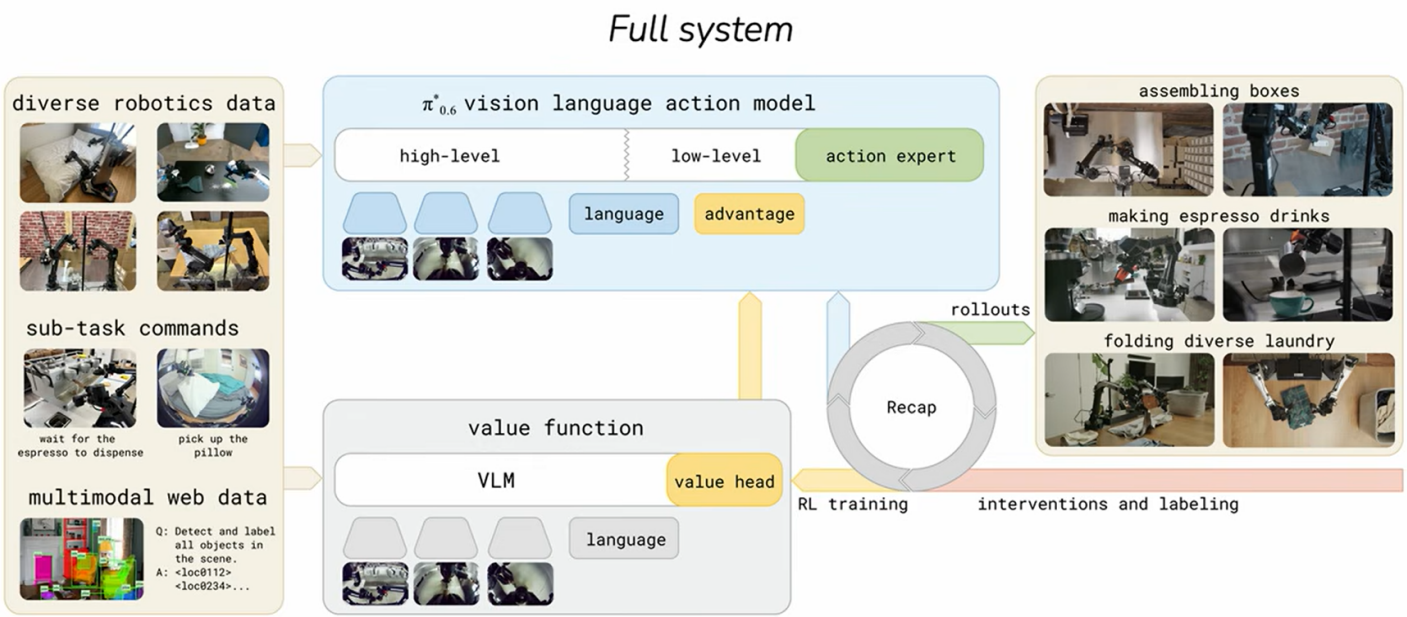
Uncertainty



Sources of uncertainty:

- **Perception** – noisy, partial sensor readings (*partial observability*)
- **Action** – actuators slip, grasps fail (*stochastic outcomes*)
- **Model** – dynamics and environment only approximately known

VLA Structure

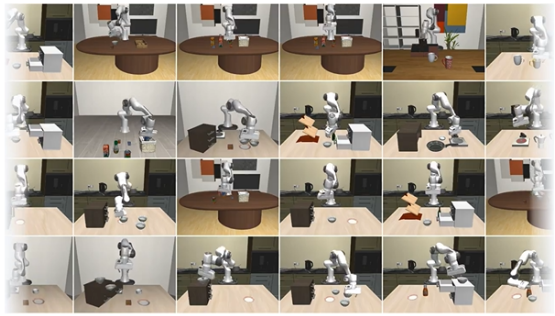


Why do we still need to understand robotics?



What are included

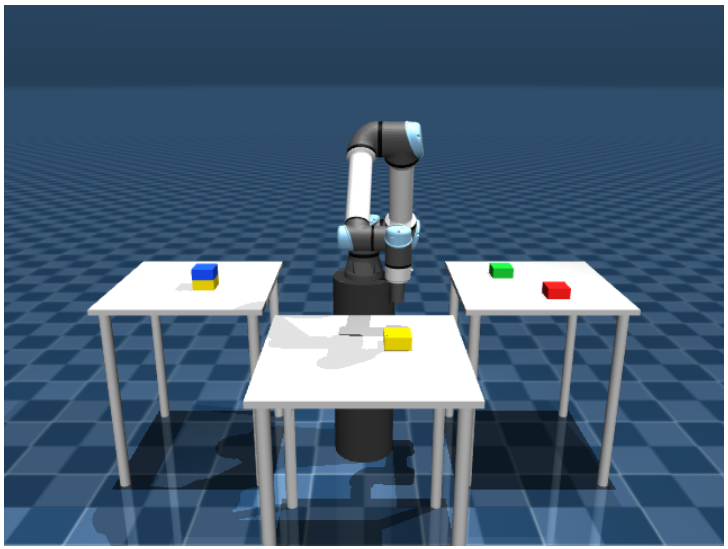
✓ RGB images from workspace and wrist cameras
✓ Proprioception
✓ Language task specifications
✓ PDDL scene descriptions



Reason, plan, act

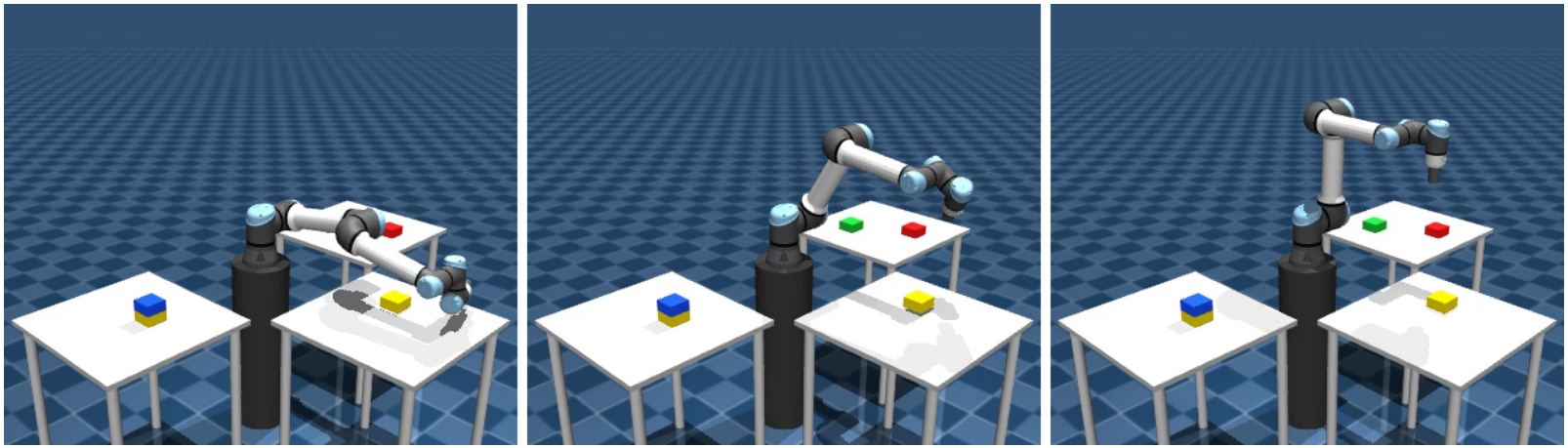
Robots that carry out multi-step tasks need two kinds of reasoning:

- **Task planning**: which actions, in which order?
  (symbols: blocks, tables, `pick-up`)
- **Motion planning and control**: how to move to carry them out?
  (numbers: joint angles, torques)



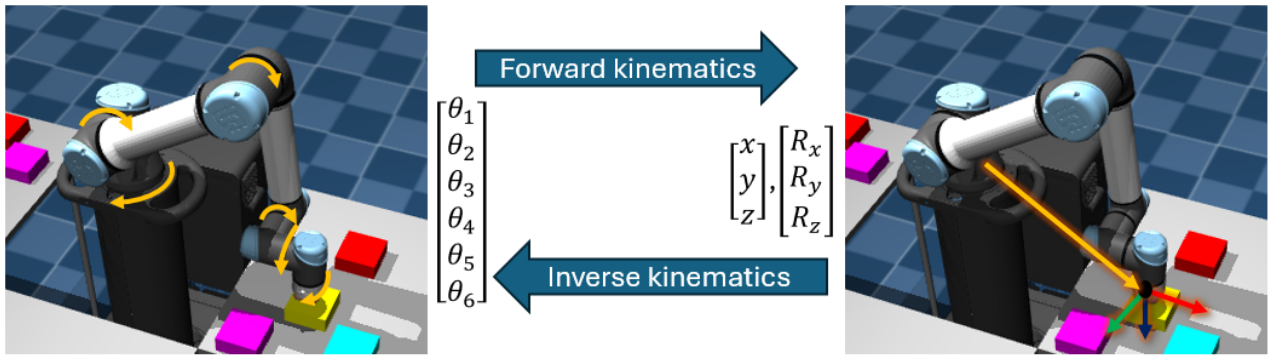
This lab is about the first.

Moving the robot



- Our **actions are target configurations**: `data.ctrl[ctrl_idx] = target`
- Position servos push each joint toward its target
- `mj_step` advances the physics by 2 ms. We keep the target applied, and physics decides what happens on the way

Forward and inverse kinematics



Forward: a chain of homogeneous transformations $T = \begin{bmatrix} R & t \\ 0 & 1 \end{bmatrix}$

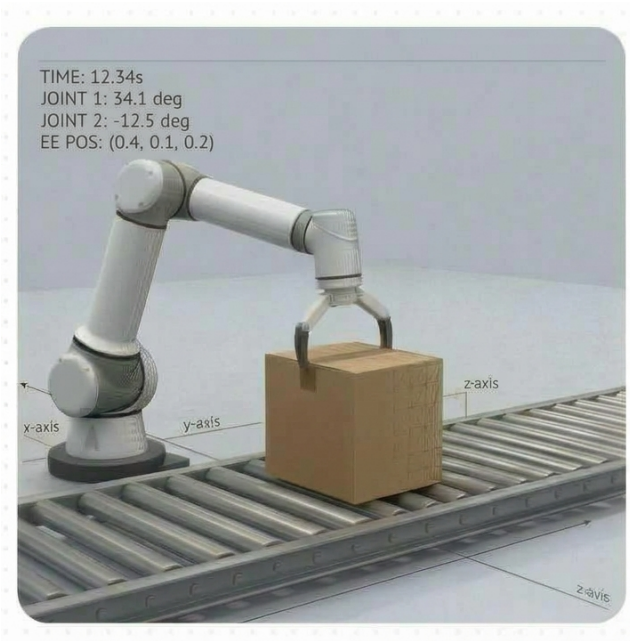
$$T_{\text{tcp}}(q) = T_{\text{base}}\, T_1(q_1)\, T_2(q_2) \cdots T_6(q_6)\, T_{\text{tool}}$$

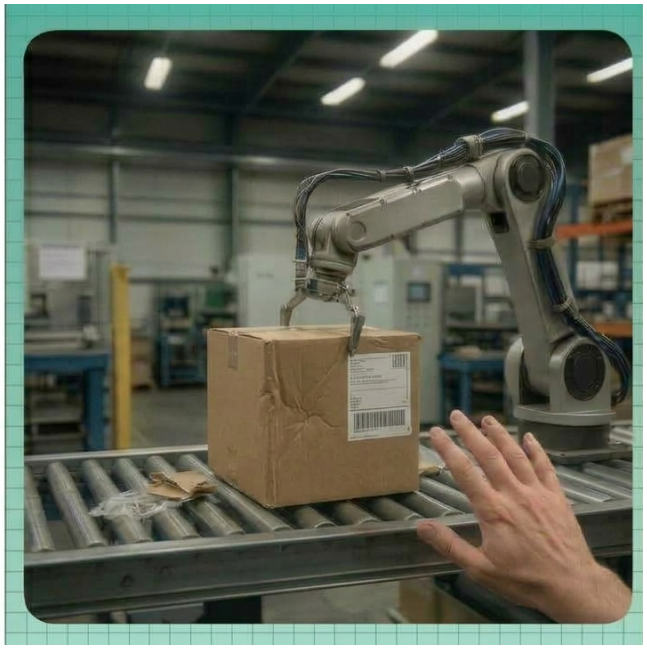
MuJoCo computes it in `mj_forward`.

Inverse is harder: the map is nonlinear, and a pose may have no solution, several, or infinitely many.

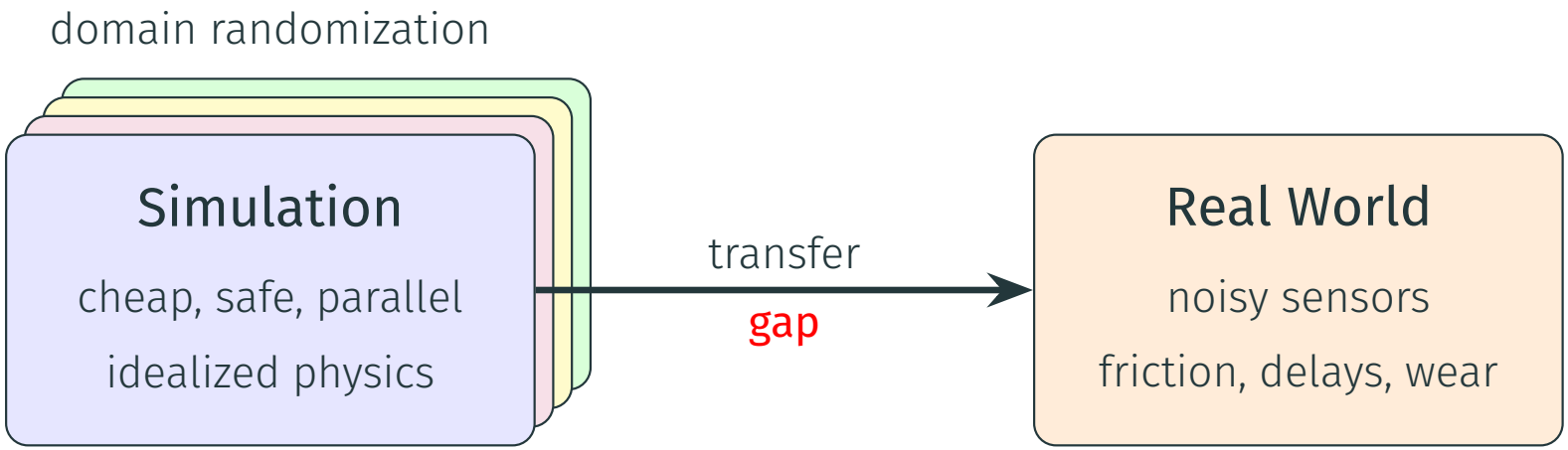
Challenges

Sim-to-Real Gap





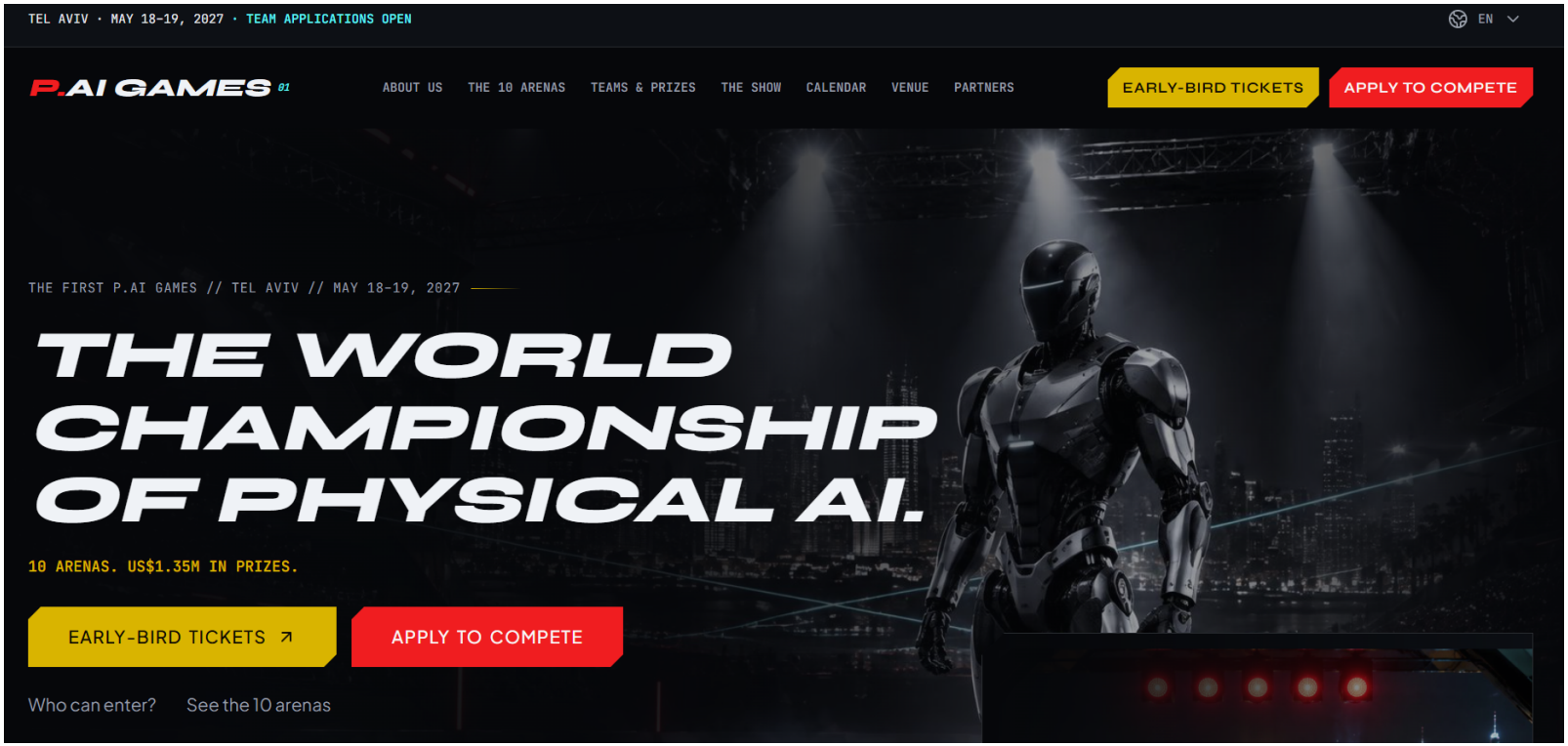
Sim-to-Real Gap



Causes: dynamics mismatch, rendered vs. real images, unmodeled effects (latency, wear)

Fixes: domain randomization, system identification, domain adaptation, real-world fine-tuning

PAI Games



https://pai-games.com/